

1.2: Polynomial Functions

A **polynomial function** is made up of several terms written in descending orders of x . The general form is:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0$$

where: n is a whole number
 x is a variable
 $a_0, a_1, \dots, a_n$ are coefficients
 n is the degree (largest exponent)
 a_n is the leading coefficient
 a_0 is the constant term

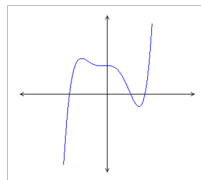
Notes:

- The power function $f(x) = ax^n$ is also a polynomial function.
- The function $f(x) = a_0$ is a polynomial function where $n = 0$, a_0 is a constant, and $f(x)$ is a horizontal line.

example: $f(x) = 10x^4 - 3x^2 + 7x - 5$

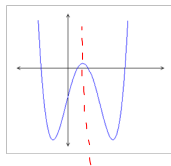
Examples:

a) $y = 3x^5 - 6x^3 - 2x^2 + 4$



Notice that the graph extends from Q3 to Q1 but NO POINT SYMMETRY exists.

b) $y = x^4 - 4x^3 - 2x^2 - 12x - 6$



Notice that the graph extends from Q2 to Q1 and the graph has LINE SYMMETRY but not in the y -axis.

Ex 1: Identify which of the following are polynomial functions.

a) $y = -3x^5$ b) $y = x^4 + 3x^3 - 5x - 1$

c) $f(x) = \cos x$ d) $y = 2x^5 + x^4 - 5x^2 - 3$

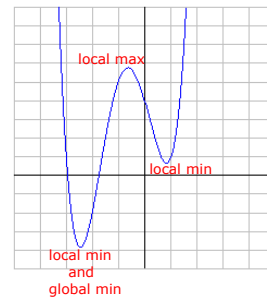
Notes when graphing polynomial f'ns:

Even-degree polynomial f'ns also have end-behaviours that extend from Q2 to Q1 *or* Q3 to Q4 (just like even power functions), but do *not necessarily* have line symmetry.

Odd-degree polynomial f'ns also have end-behaviours that extend from Q3 to Q1 *or* Q2 to Q4 (just like odd power functions), but do *not necessarily* have point symmetry.

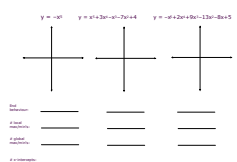
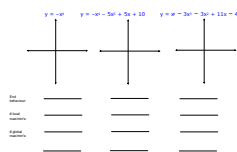
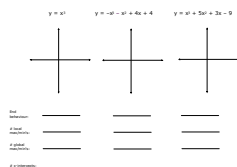
Global max/min: the absolute highest and/or lowest point on a graph

Local max/min: a hill/valley highest/lowest point on an interval



Investigation 1: What are the key features of the graphs of polynomial functions?

Sketch each function (using a smartphone or a graphing calculator). Compare the graphs. Describe their similarities and differences in terms of end behaviour; number of local/global maximum points and minimum points (see page 15 for definition); and number of x-intercepts.



Investigation 2:

Polynomial Functions and their Finite Differences

On a separate sheet of paper, construct a finite differences table for each function. Extend the table until you get a constant difference.

1. $y = x^3$

x	y	1st	2nd	3rd	4th
-3	-27				
-2	-8				
-1	-1				
0	0				
1	1				
2	8				
3	27				
4	64				

2. $y = -2x^4$

x	y	1st	2nd	3rd	4th
-3					
-2					
-1					
0					
1					
2					
3					
4					

What is true about the third differences of a cubic function?

What is true about the fourth differences of a quartic function?

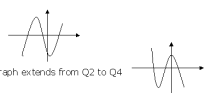
How is the sign of the leading coefficient related to the sign of the constant value of the finite differences?

How is the value of the leading coefficient related to the constant value of the finite differences?

1.2: Characteristics of Polynomial Functions Summary Sheet

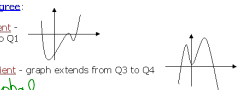
Polynomial Fns with ODD Degree:

- **Positive Leading Coefficient** - graph extends from Q3 to Q1
- **Negative Leading Coefficient** - graph extends from Q2 to Q4
- May have from a minimum of **one** up to a maximum of **n** x-intercepts (where n is the degree of the fn)
- Domain is $\{x \in \mathbb{R}\}$
- Range is $\{y \in \mathbb{R}\}$
- Have NO maximum/minimum point
- May have point symmetry (but not always)



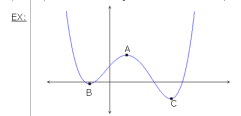
Polynomial Fns with EVEN Degree:

- **Positive Leading Coefficient** - graph extends from Q2 to Q1
- **Negative Leading Coefficient** - graph extends from Q3 to Q4
- Will always have a **global** maximum/minimum point
- Domain is always $\{x \in \mathbb{R}\}$
- Range is either $\{y \in \mathbb{R} \mid y \leq a\}$, or $\{y \in \mathbb{R} \mid y \geq a\}$, where a is the maximum/minimum value of the fn
- May have from a minimum of **zero** up to a maximum of **n** x-intercepts (where n is the degree of the fn)
- May have line symmetry (but not always)



For ALL Graphs of Polynomial Functions:

- May have at most a combined total of **n - 1** LOCAL MINIMUM and LOCAL MAXIMUM points (where n is the degree of the fn and $n > 1$, $n \in \mathbb{N}$).



Point A is a **LOCAL** maximum.

Points B and C are **LOCAL** minimums.

Point D is also considered to be an **absolute** or **global** minimum because the graph will never extend below this point.

This function does not have an **ABSOLUTE** or **GLOBAL** maximum point since there are points that extend beyond A.

For ALL Graphs of Polynomial Functions:

The n^{th} differences:

- Are equal (constant)
- Have the same sign as the leading coefficient
- Are always equal to: **$n!$** (read as: **a times "n factorial"**)
which is the same as: **$a[n \times (n-1) \times (n-2) \times \dots \times 2 \times 1]$**
where: **a** is the leading coefficient and n is the degree of the polynomial
- For example, given the polynomial function $y = -3x^5 + 4x^3 - x^2 + 7$, the 5th differences will be constant and equal to:

$$\begin{aligned} n! &= -3(5!) \\ &= -3(5 \times 4 \times 3 \times 2 \times 1) \\ &= -3(120) \\ &= -360 \end{aligned}$$

HOMEWORK: p. 11# 1, 2; p. 19# 1; p. 26# 1 - 6